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# GKN-Stack: An ensemble deep learning framework for blood glucose forecasting based on medical examination data

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**ABSTRACT** **Objective**: Diabetes patients are closely related to blood glucose levels. Predicting blood glucose levels through routine blood test data can provide auxiliary diagnosis for diabetes risk prediction in the medical field. However, physical examination datasets are often accompanied by problems such as high feature dimensions and uneven blood glucose distribution, which significantly affect the effect of machine learning models. **Methods**: This paper proposes a GA-KDE-GAN stacking model combined with feature engineering technology, referred to as the GKN framework. GKN integrates genetic algorithm and random forest (GA-RF) for feature selection, kernel density estimation (KDE) for data smoothing and small sample oversampling, and generative adversarial network (GAN) for expanding the training set. The framework uses GA-RF to select feature subsets and obtain the global optimal solution based on LightGBM evaluation, and applies KDE and GAN to balance the dataset. The final model adopts a stacking strategy to enhance the accuracy of blood glucose prediction. **Results**: By combining GKN feature engineering, the proposed model showed significant performance improvement. Under the challenging data high dimensionality and complexity, the model achieved a mean square error (MSE) of 1.529 and the highest R-square. More importantly, it significantly improved the accuracy of diabetes classification, with accuracy (Acc) and precision (Pre) exceeding 97%. **Conclusion**: This study addressed the problem of high feature dimension and uneven sample distribution in the physical examination dataset. The GKN framework proved to be effective in improving the prediction performance by integrating GA-RF, KDE and GAN. These findings are promising for glucose-assisted diagnosis of diabetes, as they can predict blood glucose levels based on routine blood test data and help in diabetes risk assessment.

**INDEX TERMS** Blood glucose estimate, feature engineering, machine learning, stacking, data augmentation.

## I. INTRODUCTION

THE advances in medical technology have led to the development of sophisticated complementary diagnostic tools that have greatly improved the accuracy of disease diagnosis. In particular, these tools are essential to addressing the challenges posed by an aging global population and the rising incidence of chronic diseases such as diabetes. Unlike traditional diagnostic methods, modern complementary diagnostic technologies can delve deeper into biomarker analysis to precisely differentiate between diseases with similar symptoms [1–3]. In diabetes diagnosis, for example, these technologies can identify subtle changes in blood sugar levels, providing early detection and reducing the likelihood of misdiagnosis in complex cases [4, 5]. Under the trend of global population aging, many chronic diseases are increasingly ravaging people. Among them, Diabetes Mellitus (DM) and the chronic diseases it causes are becoming increasingly serious global health and safety issues, not only causing the deaths of 422 million adults, but also causing 3.7 million deaths each year [6]. On December 6, 2021, the International Diabetes Federation (IDF) official website released the 2021 Global Diabetes Map. The data showed that the number of



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adult diabetes patients in the world reached 537 million in 2021, affecting about 1 in 10 adults worldwide. Compared with 2019, the number of diabetes patients increased by 74 million, an increase of 16%. It is estimated that by 2025, the prevalence of diabetes will further increase to 12.2%, and the number of patients will increase to 783 million. In addition, diabetes causes nearly 1 trillion US dollars in medical and health expenditures worldwide, accounting for 9% of total medical and health expenditures [7]. It can be said that diabetes not only brings serious health problems to patients, such as cardiovascular disease and kidney disease, but also brings heavy economic burdens to society and families. However, with current technology, diabetes cannot be cured, and its complications are usually highly stubborn, so early prevention and intervention of diabetes are of great significance [8].

According to the American Diabetes Association, the diagnosis of diabetes can be based on blood glucose standards, namely: glycated hemoglobin ($\geq$ 6.5%), fasting blood glucose ($\geq$ 7.0 mmol/L), OGTT (2-hour value $\geq$ 11.1 mmol/L), and random blood glucose $\geq$ 11.1 mmol/L in patients with typical hyperglycemia or severe hyperglycemia symptoms [9]. Based on the above standards, blood glucose prediction has become an important means to assess the risk of diabetes. As people's material life becomes richer and they pay more attention to their health status, regular blood tests have become a long-term habit for people. However, blood glucose tests are generally not included as an examination indicator in routine pre-employment physical examinations and community physical examinations. Predicting blood glucose levels from routine blood test data has important practical value, especially when frequent blood glucose monitoring is impractical. On an individual level, people with diabetes or at risk of developing diabetes can benefit from regular comprehensive blood testing, which may reduce reliance on continuous glucose monitoring (CGM) devices. This approach can enhance early detection of abnormal blood glucose, help better manage the disease, and guide treatment adjustments during routine physical examinations. For society, the risk factors for diabetes vary from region to region. It is necessary to implement specific public health policies and large-scale screening of people at risk of diabetes based on local characteristics. For society, the risk factors for diabetes vary from region to region. It is necessary to implement specific public health policies and large-scale screening of people at risk of diabetes based on local characteristics [42]. By predicting and estimating blood sugar levels based on physical examination data in blood test samples, community populations can be screened for diabetes, thereby achieving the goal of early detection and early treatment of potential patients within the scope of secondary prevention [10]. Therefore, how to predict blood sugar levels from routine blood test data is of great research significance.

The development of modern information technology and data analysis technology, such as data mining and machine learning, has provided new possibilities for the development and optimization of diabetes risk prediction models. It has been proven to be effective to improve the specificity and sensitivity of diabetes diagnosis by performing data mining and machine learning on large-scale blood test data and blood glucose levels [11]. However, routine blood testing for blood glucose prediction faces three major challenges:

- Routine blood tests cover a wide range of biomarkers, resulting in physical examination data generally having high attribute dimensions, many of which are not directly related to glucose metabolism, thus diluting the ability to predict blood glucose levels.
- Physical examination data sets are often accompanied by the problem of unbalanced intervals of blood glucose values, which causes the model to have a large prediction deviation for samples in the hypoglycemia and hyperglycemia intervals, and the generalization ability is reduced.
- The sample size of physical examination data sets is often insufficient, resulting in weak generalization and prediction capabilities of a single predictive regression model.

To address the above issues, we proposed a novel method to enhance diabetes prediction using blood glucose data obtained from routine blood tests. The problems of high attribute dimensionality, unbalanced blood glucose intervals, and insufficient sample size were addressed using an integrated stacked learning model. Specifically, our method combines Genetic Algorithm-based Random Forest (GA-RF) for feature selection, Kernel Density Estimation (KDE) for resampling, and Generative Adversarial Network (GAN) for data augmentation. Finally, a stacking ensemble strategy was used to improve prediction accuracy and generalization ability. And in order to avoid the Genetic Algorithm from falling into a local optimal solution, we introduced a Light Gradient Boosting Machine (LightGBM) based on a tree-based learning algorithm to evaluate feature subsets. The combination of these GA-KDE-GAN (GKN) techniques enables us to overcome the limitations of previous methods and achieve more reliable predictions. The contributions of this paper are as follows:

1) A GA-RF feature selection algorithm based on LightGBM as a suboptimal attribute subset evaluation model is proposed to extract the optimal attribute subset of preprocessed data. This method effectively solves the problems of too high attribute dimension and too many useless features in the data set, thereby improving the training efficiency and prediction accuracy of the model.
2) The KDE technology is combined with the GAN to estimate the probability density function of the data distribution, resample the imbalanced data set, and expand the data set. This combined data enhancement strategy balances the sample distribution, enriches the diversity of training data, and improves the robustness and generalization ability of the model.
3) A stacked ensemble learning model consisting of four



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models is proposed. Cluster label enrichment features are introduced to help the model learn the potential distribution of training data, and the data set is trained under a five-fold cross-validation training strategy. Experiments show that the model has better generalization ability and better prediction performance.

The paper's structure is delineated as follows: Section II provides a summary of previous literature concerning Diabetes Prediction. Section III encompasses details about the dataset, experimental methodologies, and the proposed approach. Section IV delivers the achieved results, and Section V encapsulates the conclusion while outlining potential avenues for future research.

## II. RELATED WORK

Combining data mining, IoT, and machine learning to analyze medical data has become a potential solution, and ML models have also been shown to be feasible in identifying early risks associated with DM.

Current machine learning research on predicting diabetes risk mainly predicts and classifies diabetes by analyzing early symptoms of diabetes. Dong et al. [12] used machine learning methods to construct a 3-year diabetic nephropathy risk prediction model for patients with type 2 diabetes, and compared the prediction performance of seven machine learning algorithms, including LightGBM, XGBoost, and SVM, and concluded that LightGBM performed best in the result of experiment. Rupapara et al. [13] studied an ensemble learning method for diabetes detection using the PIMA India dataset. They compared PCA, Chi-2, and original features and found that Chi-2 features performed best. The proposed three-integrated model achieved an accuracy of 85% in predicting diabetes. Alpan et al. [14] used the WEKA tool to conduct diabetes data mining analysis on the UCI data set, and compared seven basic machine learning algorithms through experiments. Among them, the accuracy of the KNN algorithm is the highest after using 10-fold cross-validation technology, reaching 98.07%. With the continuous advancement of deep learning research, it has been increasingly used in the prediction of diabetes. M. Arora et al. [15] used deep (DNN) technology and achieved an accuracy of 74.4% through CNN. Madan et al. [16] proposed to merge two models CNN-Bi-LSTM to detect (and predict) type 2 diabetes. Experiments showed that CNN-Bi-LSTM surpassed other deep learning methods in accuracy (98%), sensitivity (97%) and specificity (98%). Rahman et al. [17] used the diabetes risk prediction model based on Conv-LSTM for diabetes risk prediction for the first time. They compared the performance of Conv-LSTM, T-LSTM, CNN and CNN-LSTM on the Indian PIMA diabetes dataset, among which Conv-LSTM outperformed the other three models in terms of classification accuracy.

In the prediction problems related to diabetes, a single classification or prediction model generally has problems such as low accuracy, poor generalization ability, and poor fault tolerance. Ensemble learning can combine multiple inducing factors or multiple learners to make decisions [18]. This method has a certain improvement in generalization ability and accuracy, and is the development trend of studying diabetes prediction problems. In terms of early prediction of diabetes, Dutta et al. [19] proposed a projection weighted ensemble method (combining random forest, decision tree, LightGBM and XGBoost). This method combined with preprocessing technology can significantly improve the prediction performance with an accuracy of 73.5%. Ihnaini et al. [20] used an integrated deep learning model and data fusion technology for diabetes prediction. Compared with other classifiers, this method has the best performance, achieving an accuracy of 99.64%, an accuracy of 100%, and a sensitivity of 99% on the fused data set. Taha et al. [21] proposed an ensemble learning method based on a hybrid meta-classifier of fuzzy clustering and logistic regression for predicting type 2 diabetes. This method performed better than 6 individual classifiers and 2 ensemble methods.

Diabetes prediction data sets based on physiological drivers often face the characteristics of high feature dimension and many useless features, which have a great impact on subsequent training. Therefore, feature selection of the data set is required [22, 23]. Deberneh et al. [24]used variance analysis, chi-square test and recursive feature elimination method to select 12 feature indicators from high-dimensional features, and compared the prediction performance of each classifier based on this. In addition to traditional feature elimination methods, genetic algorithms based on input packaging methods can successfully extract attribute subsets from high-dimensional feature sets, speeding up model training while improving prediction accuracy [25]. Li et al. [26] used a combination of Harmony search algorithm, Genetic Algorithm (GA) and Particle Swarm Optimization (PSO) with K-means clustering algorithm for feature selection, and used K-nearest neighbor (KNN) to classify the diabetes dataset, achieving an accuracy of 91.65%. Tan et al. [10] proposed an integrated learning model(genetic algorithm-stacking, GA-stacking) that combines genetic algorithm for feature selection. Through comparative experiments of KNN, SVM, LR, NB, CNN and stacking models on the data set , indicating that adding GA can improve the prediction efficiency and performance of each model, and GA-stacking has the highest average accuracy, sensitivity, and specificity on the Qingdao physical examination data set, which are 85.88%, 99.24%, and 99.92% respectively.

There are few studies on diabetes classification using physical examination datasets, but still related studies in Reference [27] and Reference [10], but there are even fewer studies on blood glucose prediction. Although the above research results have high performance, they also have some limitations. For example, most of the studies are based on the classic PIMA Indian diabetes dataset and the UCI diabetes dataset, which may not be as effective in predicting regression problems as classification problems. In addition, since physical examination datasets are usually high-dimensional and have unbalanced blood glucose intervals, they are obviously more challenging than traditional diabetes classification problems.



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However, whether it is diabetes classification or blood glucose prediction problems, further research is needed to improve the performance of the model in diabetes detection, including exploring innovative architectures, feature selection, data augmentation, and parameter adjustment. A summary of pertinent studies can be found in Table 1.

## III. MATERIALS AND METHODS

### A. DIABETES DATASET

The dataset used in this paper comes from 5112 sample data of the Alibaba Cloud Tianchi Precision Medicine Competition dataset [28]. Each data in the dataset contains 41 physical examination information of the user, such as age, hemoglobin, white blood cell ratio and other single examination items, and the target variable is blood glucose value, and its blood glucose density distribution is shown in Figure 1. Since some features of the dataset contain a large number of missing values, even if the interpolation method is used to fill the samples, it is easy to introduce unnecessary noise. Therefore, when training the dataset, it is necessary to delete the attribute columns with serious missing values and the samples with a large number of missing attribute column data. After deleting the missing attribute columns, the target dataset contains 32 attribute features. Figure 2 shows the distribution histograms of some features in the training set, as well as the corresponding distribution box plots of these features. The histogram intuitively shows the distribution of the data, while the box plot can intuitively show the central tendency and outlier distribution of the data. Due to the large differences in the distribution of features, the data set needs to be normalized before training the model.

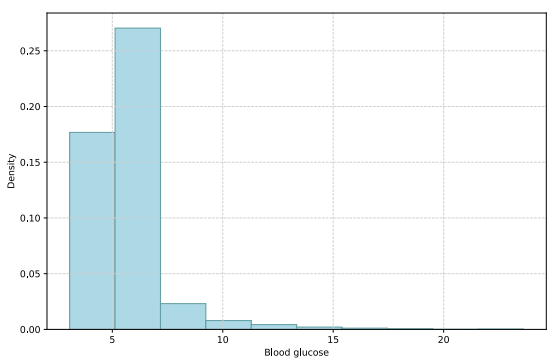



FIGURE 1: Density distribution of blood glucose levels.

In addition, in order to verify the performance of the proposed model in the diabetes risk classification problem, this paper also selected the fusion dataset in the Reference [20] for experiments. The Hospital Frankfurt Germany diabetes dataset [29] consists of 2000 cases with eight features. The PIMA Indians diabetes dataset [30] consists of 768 patients with 8 features. The fusion feature dataset used for the experiment is a combination of the features of the two datasets. There are 2768 case data with eight features, and some of the missing values of the dataset cases are filled by averaging. Detailed features description of fused features is shown in Table 2.

### B. FEATURE SELECTION BASED ON GA-RF AND LIGHTGBM

Feature selection methods are mainly divided into filtering, wrapping and embedding. Because there is a certain relationship between the features of the dataset studied in this paper, wrapping and embedding are suitable. The embedding method generally uses some interpretable machine learning methods [31, 32] to understand the importance of each feature in the dataset to the prediction result. However, for nonlinear relationships or high-order interactions, the importance evaluation of a single feature may not provide enough information. Feature selection based only on feature importance ranking will be affected by the prediction error and noise characteristics of the model itself. Therefore, this paper uses a self-inspired genetic algorithm for feature selection.

Although the genetic algorithm has the ability to search globally, it may actually fall into a local optimal solution due to improper population initialization. The simplest way to solve this problem is to run the algorithm multiple times and iterate multiple times. Therefore, this paper introduces lightgbm to evaluate whether the feature subset is optimal. The alternating evaluation of lightgbm and random forest can make the selected feature subset more reliable and greatly increase the probability that the target subset is the global optimal solution.

In order to better understand the correlation between attribute features in the target dataset, we drew a feature heat map, as shown in the Figure 3. When the color leans toward blue, the correlation between features is lower; when the color leans toward red, the correlation is higher. It can be seen from the figure 3 that the redundancy between most attributes is small, but the redundancy between a small number of attributes is relatively large. To select the optimal representative attribute subset, this article uses genetic algorithm combined with random forest (GA-RF) for feature selection, and introduces the LightGBM model for evaluation and iterative update to ensure the optimality of the attribute subset.

Since the physiologically driven blood glucose prediction data set involves a large amount of physical data and the feature dimension is too large, it is not suitable for direct use in blood glucose prediction regression. For the original blood glucose data set with large feature dimensions, feature selection is required to reduce the noise interference of useless features and reduce the probability of over-fitting. In addition, the initialization population of the genetic algorithm is random. The attribute subset obtained by feature selection based on the decision tree genetic algorithm is not necessarily optimal or unique. In order to solve this randomness, it is necessary to introduce an evaluation strategy into the feature selection algorithm for iterative training. This paper uses a genetic algorithm based on a combination of input packing



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TABLE 1: Overview of related work.

| Reference | Year | Models | Database | Report result |
|---|---|---|---|---|
| [13] | 2023 | AdaBoost, GNB, DT, RF, LTC | PIMA Indian dataset | 85% (LTC) |
| [14] | 2020 | BN, NB, J48, RF, KNN, SVM | UCI diabetes dataset | 98.07% (KNN) |
| [16] | 2022 | CNN, Bi-LSTM, CNN-Bi-LSTM | PIMA Indian dataset | 98% (CNN-Bi-LSTM) |
| [17] | 2020 | Conv-LSTM, T-LSTM, CNN, CNN-LSTM | PIMA Indian dataset | 97.26% (Conv-LSTM) |
| [19] | 2022 | NB, RF, DT, XGboost, LightGBM, Ensemble(RF, DT, XGboost, LightGBM) | diabetes dataset from Bangladesh | 73.5% (Ensemble) |
| [20] | 2021 | NB, RF, DT, KNN, SHRS-M3DP model | Fused Dataset of PIMA Indian and Hospital Frankfurt Germany diabetes | 99.64% (SHRS-M3DP model) |
| [24] | 2021 | feature selection: VA, CST, RFE;<br>models: LR, RF, SVM, XGboost, Ensemble(LR, RF, SVM, XGboost) | diabetes dataset from Hanaro Medical foundation in Seoul, South Korea | 81% (Ensemble) |
| [26] | 2021 | Ensemble feature selection: (HSA, GA and PSO);<br>model: KNN | PIMA Indian dataset | 91.65% (KNN after feature selection) |
| [27] | 2021 | DT, RF, SVM, NB, LR | Physical examination electronic medical record dataset | 94.2% (RF) |
| [10] | 2022 | feature selection: GA-RF;<br>model: KNN, LR, SVM, NB, CNN, GA-stacking | Qingdao Physical Examination Dataset;<br>UCI diabetes dataset | 85.88% (GA-stacking on Physical Examination Dataset);<br>98.71% (GA-stacking on UCI diabetes dataset) |

TABLE 2: The fusion diabetes dataset description.

| no. | Attribute | Unit | Ranges |
|---|---|---|---|
| 1 | Age | year | 21-81 |
| 2 | Pregnancies | Number | 0-17 |
| 3 | Glucose | mg/Dl | 44-199 |
| 4 | Blood Pressure | mm Hg | 90-122 |
| 5 | Skin Thickness | mm | 0-110 |
| 6 | Insulin | uU/ml | 0-744 |
| 7 | BMI | Kg/m2 | 18.2-80.6 |
| 8 | DBF | Number | 0-1 |
| 9 | outcome | Positive (1), negative (0) | 0, 1 |

method and random forest, supplemented by LightGBM as an evaluation model to achieve the acquisition of the optimal attribute subset. The flow chart of the entire feature selection algorithm is as follows in figure 4: Firstly, the population is initialized. The population contains multiple individuals. The characteristic value of each individual is randomly selected from 0 or 1. Secondly within a given number of iterations, new individuals are generated through operations such as crossover and mutation, and the values are calculated according to the fitness function. They evaluate and then update the feature vectors of individuals in the population in order to obtain the best feature subset of the population. Then based on this feature subset, the LightGBM model is used for multiple training and evaluation, and a better attribute subset is recorded at the same time. At last after a certain number of iterations, the optimal feature subset of the original dataset is finally obtained.

### C. EXCESSIVE SAMPLING BASED ON KDE ESTIMATION

In the concentration of blood glucose data, high blood glucose and hypoglycemia samples are usually imbalance. Directly using these unbalanced data training models may cause predicts of model prediction for minority samples. Because GAN usually needs a large amount of data for effective training, and the high and low blood glucose samples are insufficient, this challenge the use of GAN network for small samples. In order to balance the dataset, we use Nuclear Density Estimation (KDE) to estimate the probability density function of each sample type, as well as re-sampling based on this. KDE is a non-parameter method that is used to estimate the probability density function of random variables. Compared with GAN, the KDE computing complexity is low and does not require the preset data distribution form. It can flexibly adapt to the distribution of various practical data. The new sample generated can reflect the true distribution of the original data and the quality is high.

In this article, we are divided into hypoglycemia, normal blood glucose, and hyperglycemia according to the blood glucose range. Using KDE with KDE with less data and hypoglycemia samples for less data volume, the probability density function is estimated to get the probability distribution of these two types of samples:

$$\hat{f}_{\text{high}}(x) = \frac{1}{n_{\text{high}}\, h} \sum_{i=1}^{n_{\text{high}}} K\left(\frac{x - x_{\text{high},\, i}}{h}\right) \tag{1}$$

$$\hat{f}_{\text{low}}(x) = \frac{1}{n_{\text{low}}\, h} \sum_{i=1}^{n_{\text{low}}} K\left(\frac{x - x_{\text{low},\, i}}{h}\right) \tag{2}$$



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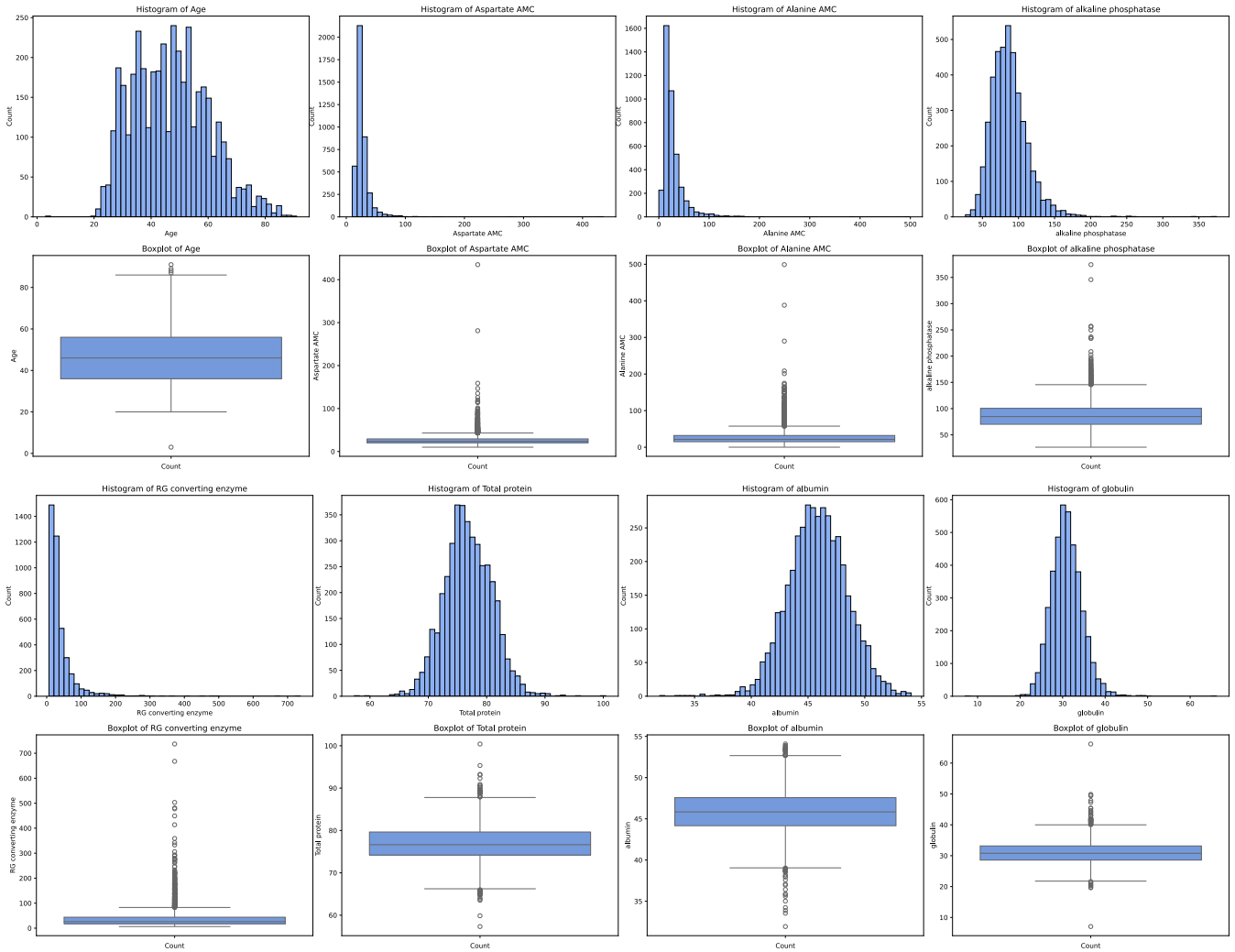



FIGURE 2: The distribution histograms of some features in the training set, as well as the corresponding distribution box plots of these features. Because the distribution differences between features are large, the dataset needs to be normalized before training the model.

where, $K$ is the kernel function, the commonly used kernel function is Gaussian kernel, $h$ is the bandwidth parameter, which controls the smoothness.

According to the estimated probability density function, samples are randomly selected from high and low blood glucose samples, and the generated samples are added to the original blood glucose sample set for data set reconstruction.

### *D. DATA AUGMENTATION BASED ON GENERATIVE ADVERSARIAL NETWORKS*

Due to the complexity of the model, model stacking and cross-validation strategies place certain requirements on the sample size of the dataset. The original target dataset used in this article has a small sample size. Therefore, in order to increase the diversity of training data, this paper proposes a data enhancement method based on generative adversarial networks(GAN) [33], which trains the generator network to generate features similar to new samples of the original training data.

As shown in the Figure 5, GAN data enhancement involves two main components: Generator and Discriminator, and the interaction between them is defined by the Adversarial Loss function. The goal of the generator is to generate realistic synthetic data to fool the discriminator as much as possible, while the goal of the discriminator is to accurately distinguish between real and generated data. This paper first conducts weakly supervised training on part of the target dataset after feature selection based on the generator, generates labels for real samples and generated samples, then trains the discriminator based on real samples and generated samples, and feeds the results back to the generator to generate The controller parameters are updated iteratively. The two train each other for a certain number of rounds until their respective loss functions are minimized. When the discriminator cannot accurately



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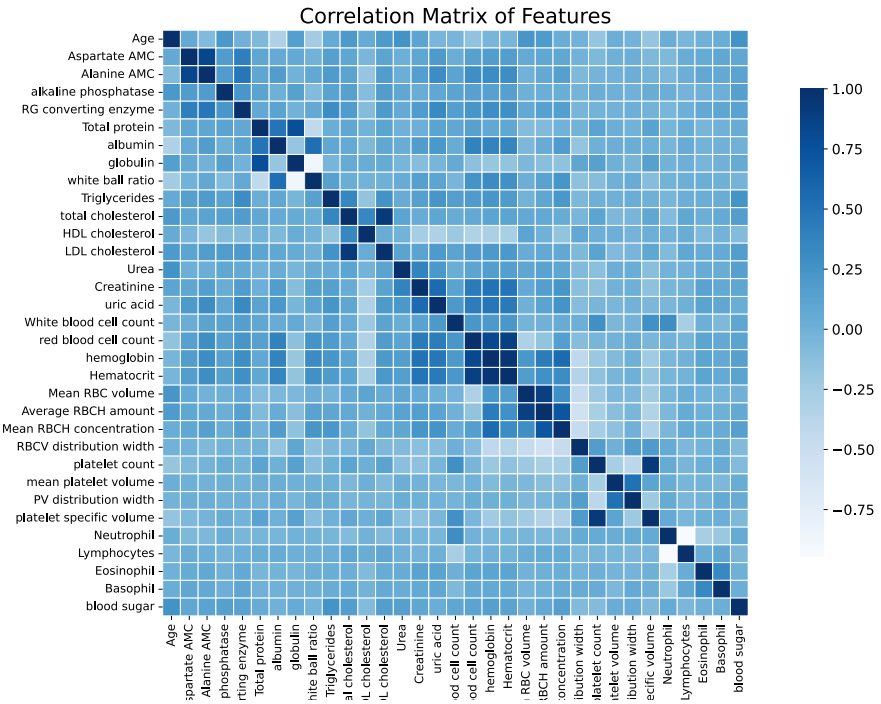



FIGURE 3: Feature correlation heatmap.

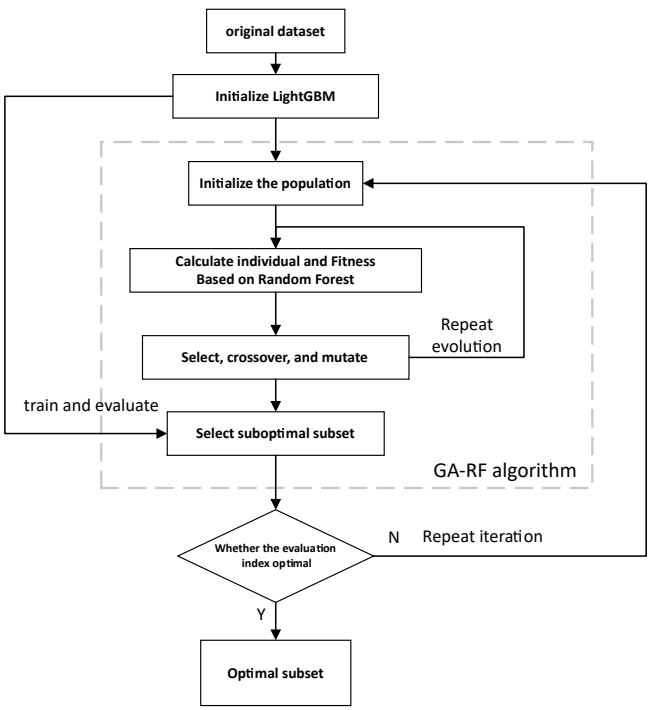



FIGURE 4: Structure diagram of GA-RF feature selection algorithm that introduces LightGBM as an evaluation model.

determine the difference between the samples generated by

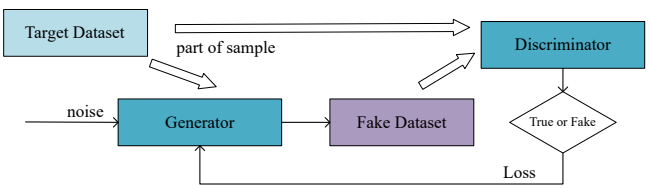



FIGURE 5: GAN network schematic diagram. The goal of the generator is to generate synthetic data that is as realistic as possible to fool the discriminator, which is responsible for distinguishing between real data and generated synthetic data.

the generator and the real samples, the generated samples can become an extension of the target dataset.

### *E. STACKING MODEL*

Stacking Model is an ensemble learning strategy proposed by David H. Wolpert [34], which aims to combine the prediction results of multiple basic models to improve the overall prediction performance. It is a meta-learning technology that takes the prediction results of different models as input and then feeds these results to another model (meta-model) to generate the final prediction result, thereby minimizing one or more generalized error rate.

The traditional stacked model mainly consists of multiple main learners and a meta-learner. Multiple main learners train data sets and generate prediction results, while the meta-learner uses the predicted output results of multiple



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main learners as input for training. Finally, five prediction results were obtained and averaged under the five-fold cross-validation strategy.

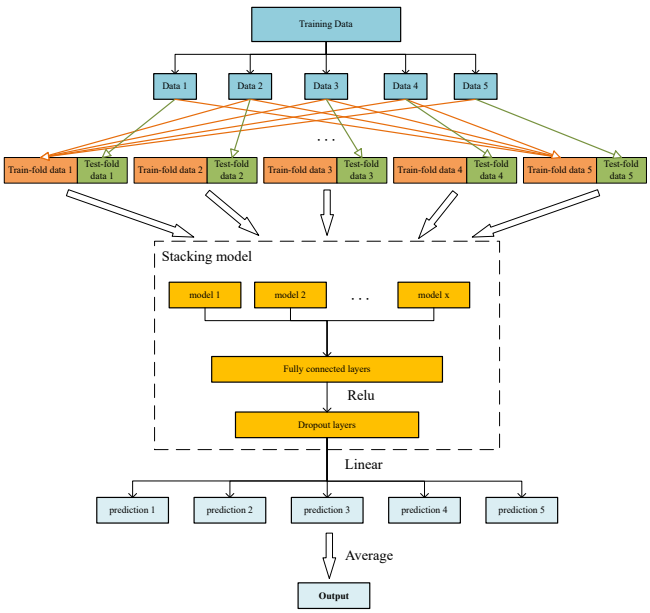



FIGURE 6: The structure of the stacking model and five-fold cross-validation strategy.

As shown in the figure 6, the stacking model used in this article first connects the prediction outputs of the four main learners and the clustering results of EM clustering through the concatenate layer to form the overall input, and then learns the potential features information of the overall input set through a series of fully connected layers and dropout layers, and finally outputs the model's predictions through a dense layer with a linear activation function. For physiologically driven blood glucose prediction data sets, regression prediction models often face the problems of limited prediction ability and weak generalization ability, and this stack can integrate the advantages of multiple predictive regression learners, as well as additional Dropout layers to reduce the risk of overfitting, making the model applicable to multiple datasets.

### *F. SUB-LEARNERS FOR STACKED MODEL*

Here, the sub-learners of the four stacked models used in this article are introduced.

#### 1) Long Short-Term Memory

Long Short-Term Memory (LSTM) is a deep learning model commonly used to process and predict time series data proposed by Sepp Hochreiter and others in 1997 [35]. It is widely used in bioinformatics, such as DNA sequence analysis, protein structure prediction, etc. It is a variant of Recurrent Neural Network (RNN) that aims to solve the problem of vanishing or exploding gradients of RNN on long sequences. It is mainly controlled by three gates: forget gate, input gate and output gate.

The three gate control units control the flow of information by using the sigmoid activation function to generate values between 0 and 1, deciding which information should be passed, forgotten or output. The input gate determines how much new information should be added to the unit state in the current time step. The forget gate determines how much past information in the current time step should be forgotten. The output gate determines how much unit state information in the current time step should be output to the next time step or as the final output of the model.The calculation formulas are expressed as:

$$i_t = \sigma(W_{xt} \cdot x_t + W_{hi} \cdot h_{t-1} + b_i) \tag{3}$$

$$f_t = \sigma(W_{xf} \cdot x_t + W_{hf} \cdot h_{t-1} + b_f) \tag{4}$$

$$o_t = \sigma(W_{xo} \cdot x_t + W_{ho} \cdot h_{t-1} + b_o) \tag{5}$$

Where $i_t$ $f_t$ and $o_t$ are the outputs of the input gate, forgetting gate and output gate respectively,$x_t$is the input of the current time step, $h_{t-1}$ is the hidden state of the previous time step, $W_x$, $W_h$ and $b$ are the weight parameters and bias terms of these gates, and $\sigma$ is the sigmoid activation function.

LSTM updates the unit state based on the decisions of the forget gate and input gate. The forget gate determines what information is retained, and the input gate determines what new information is added. The update of unit status is completed by the following formula:

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \tag{6}$$

Where $C_t$ is the unit state at time step t, $C_{t-1}$ is the unit state at the previous time step, and $\tilde{C}_t$ is the candidate value vector

#### 2) Support Vector Machine

Support Vector Machine (SVM) is a powerful supervised learning algorithm used for classification and regression problems [36]. The main idea is to effectively separate different categories of data points in high-dimensional space by finding an optimal hyperplane (or set of hyperplanes).

In regression problems, the goal of SVM is to fit the data by minimizing the loss function and control the accuracy of the fitting and the complexity of the model through the interval band. The loss function and constraints are:

$$\min_{\mathbf{w},b,\{\xi_i,\xi_i^*\}} \left\{ \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{i=1}^{n} (\xi_i + \xi_i^*) \right\} \tag{7}$$

$$\begin{aligned} y_i - (\mathbf{w} \cdot \mathbf{x}_i + b) &\leq \varepsilon + \xi_i \\ (\mathbf{w} \cdot \mathbf{x}_i + b) - y_i &\leq \varepsilon + \xi_i^* \end{aligned} \tag{8}$$

Where, $\mathbf{w}$ is the weight vector of the model, $b$ is the bias term, $\xi_i$ and $\xi_i^*$ is the slack variable and $\xi_i, \xi_i^* \geq 0$, $C$ is the regularization parameter, and $\varepsilon$ is the tolerance, which represents the allowable deviation between the predicted value and the true value.



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#### 3) Random Forest

Random Forest (RF) is one of the ensemble learning methods based on decision trees [32]. It improves the accuracy and stability of the model by constructing multiple decision trees and integrating their prediction results. Random forest introduces randomness to improve the diversity and generalization ability of the model. In a random forest, multiple training sets are generated by sampling with replacement (Bootstrap sampling) on the original data set, and then a decision tree is built for each training set. In addition, every time a node is split, instead of using all features, a portion of features are randomly selected from all features for consideration. This can reduce overfitting and increase the generalization ability of the model. Random forest has good robustness and generalization ability, can handle a large number of features and samples, and does not require too many parameter adjustments. Therefore, it is widely used in tasks such as classification, regression, and feature selection in practical applications.

#### 4) Light Gradient Boosting Machine

As an distributed gradient boosting framework based on the Gradient Boosting Decision Tree (GBDT) algorithm, developed by Microsoft [37]. Light Gradient Boosting Machine iteratively trains multiple decision trees and updates model parameters using the gradient descent method to minimize the loss function. In the regression problem, when the loss function is set to the mean square error, the objective function can be defined as:

$$obj(\Theta) = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 + \Omega(\Theta) \tag{9}$$

Where, $\Theta$ represents the parameters of the model and $\Omega(\Theta)$ represents the regularization term (used to control the complexity of the model).

Through performing Taylor expansion and simplification of the equation (8), the objective function in the following approximate form can be obtained:

$$\mathrm{obj}(\Theta, g, h) = \sum_{i=1}^{N}\left[g_i f\left(\mathbf{x}_i\right) + \frac{1}{2}h_i f^2\left(\mathbf{x}_i\right)\right] + \Omega(\Theta) \tag{10}$$

Where, $g_i$ and $h_i$ represent the first-order gradient and second-order gradient information of sample $i$ respectively, and $f\left(\mathbf{x}_i\right)$ represent the predicted value of the sample $i$.

LightGBM uses optimization methods such as Leaf-wise growth strategy, histogram difference acceleration, and parallel learning to achieve efficient, fast, and accurate training and prediction on large-scale data sets.

### G. PROPOSED GKN-STACK MODEL FOR DIABETES PREDICTION

This study uses a physical examination dataset for blood glucose prediction. The dataset is pre-processed to solve the problem of missing values by deleting some feature columns and using interpolation methods. The dataset is divided into a ratio of 8:2, 80% for model training and 20% for testing. This study proposes a blood glucose prediction and diabetes classification model GKN-Stack model. The system first uses the GA-RF algorithm to generate a set of candidate features, uses LightGBM to evaluate each subset, and selects the subset with the highest evaluation score as the optimal attribute set. Then, KDE is used to estimate the probability density function of the unbalanced data sample, and resamples on this basis to alleviate the problem of unbalanced data sets. Then, in order to meet the requirements of the stack model for the data sample size, GAN is used for data enhancement to solve the problem of insufficient data samples and increase the diversity of training data. Finally, the four algorithms are stacked using integration technology, and EM clustering is introduced to perform cluster analysis on the target dataset. The clustering labels are sent to the fully connected layer as knowledge features for learning, so that the stacked model can better identify the feature differences between samples, thereby improving the generalization ability and prediction accuracy of the model. Figure 7 describes the implementation process of this technology.

In LSTM network structure designed in this article, the input attributes are first expanded according to the input shape of the LSTM network, and then passed through two layers of hidden 36-unit and 18-unit LSTM layers, and then passed through two hidden 36-unit and 18-unit LSTM layers, and then the output of the LSTM layer is weighted and summed through the attention mechanism to capture the importance of different time steps in the target sequence. The formula of the attention mechanism is expressed as:

$$e_{ij} = \mathrm{softmax}(W_q \cdot x_i \cdot (W_k \cdot x_j)^T) \tag{11}$$

$$y_i = \sum_{j=1}^{n} e_{ij} * (W_v \cdot x_j) \tag{12}$$

where $X = x_1, x_2, ..., x_n$ is the input sequence, $e_{ij}$ is the attention weight, $y_i$ is the weighted output.

After the weighted calculation of the attention mechanism, this weighted representation is used as part of the residual connection, and is added together with the output of the LSTM as the input of the Dropout layer and the GRU layer, thereby reducing the overfitting probability of the model. Finally, the linear function of the activation layer is used to capture the nonlinear factors of the sample feature data and extract effective information.

The stacking model designed in this paper combines the features of multiple learners, takes their predicted outputs as the input of the meta-learner, trains through a fully connected layer network, and predicts the final result. We first independently train multiple different sub-learners, and the prediction results of these sub-learners will be combined into a feature matrix as a new feature set as the input of the meta-learner. The meta-model adopts a fully connected layer structure. Through training, it can identify and extract the relationship between the predictions and labels of different models. The weights obtained by training can balance the influence of each sub-learner on the final result. This method allows our



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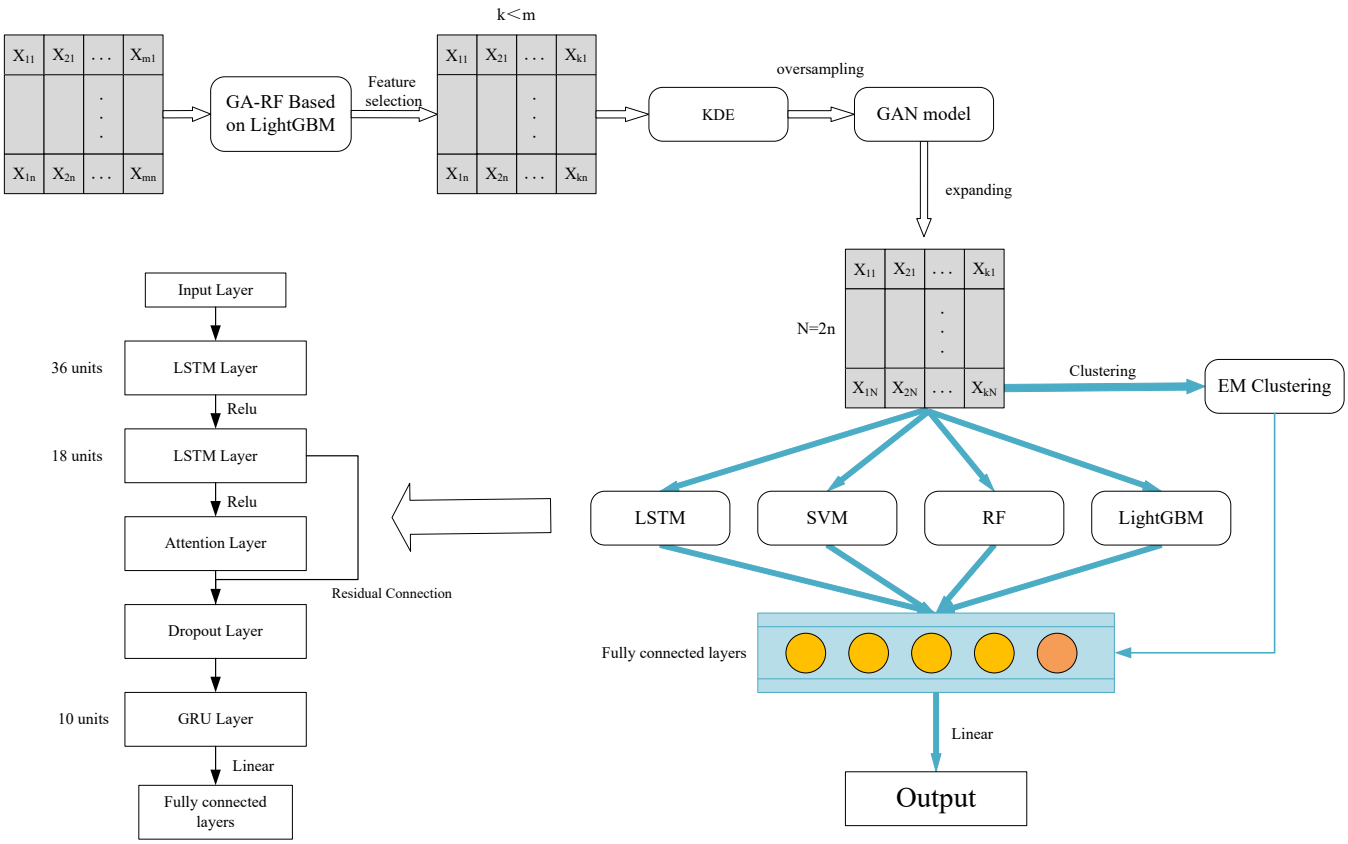



FIGURE 7: The overall structure of the proposed GKN-Stack. The training set is first evaluated by the GA-RF algorithm and LightGBM for feature selection, and then oversampled and expanded by the data enhancement method combining KDE and GAN. The integrated stacking model uses LSTM, SVM, RF and LightGBM as sub-learners, and the fully connected layer as the meta-learner. The prediction output of the sub-learner and the cluster label are used as the input of the meta-learner to improve the prediction and regression capabilities of the model. The model structure of the LSTM sub-learner is also shown.

meta-learner to pay more attention to those classifiers with more stable performance. In addition, in order to help the model better understand the structure and pattern of the data and improve the performance and robustness of the model, this paper introduces the EM clustering analysis results as knowledge features into the input layer of the stacked model to enrich the data features and better help the model extract the hidden patterns and structures between the features. The main structure of the meta-model introduces the Dropout layer and L2 regularization strategy to minimize the probability of model overfitting. At the same time, in order to improve the model's ability to recognize the data structure of abnormal blood glucose samples, we set the weight of blood glucose samples higher than 7 or lower than 4 to 2, and set the weight of blood glucose samples between 4 and 7 to 1, so that the model pays more attention to the potential feature information of unbalanced samples. The overall training strategy adopts the five-fold cross-validation method to ensure the generalization and robustness of the model.

## IV. EXPERIMENTS AND RESULTS

The software environment of the experiment is Windows 11, which is equipped with 13th Gen Intel(R) Core(TM) i7-13700H 2.40 GHz, 32G RAM, NVIDIA 4060 graphics card. The deep learning algorithms used are Keras and sklearn.

### A. EVALUATION CRITERIA

To evaluate the predictive regression effect of the stacking model and other machine learning algorithms on the blood glucose dataset, the Mean Square Error MSE), Mean Absolute Error (MAE), and R-Squared Coefficient of Determination ($R^2$) are selected as the evaluation criteria of this article. MSE is the average of the squared differences between the predicted and true values. The MAE is the average of the absolute values between the predicted and true values. The coefficient of determination is how well the model fits the data and ranges from 0 to 1. The closer the value is to 1, the better the model fits. In evaluating model predictive regression performance, the smaller the MSE and MAE, the



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better, and the closer $R^2$ is to 1, the better.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{13}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{14}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{15}$$

To evaluate the diabetes classification performance of the model, various commonly used classification evaluation metrics were used. These include Recall (Sensitivity), Precision, Accuracy, and F1-score. These metrics are calculated using a confusion matrix, which consists of the following components: true negatives (TN), false negatives (FN), false positives (FP), and true positives (TP).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{16}$$

$$Precision = \frac{TP}{TP + FP} \tag{17}$$

$$Recall(Sensitivity) = \frac{TP}{TP + FN} \tag{18}$$

$$F1 - Score(FM) = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{19}$$

### B. EXPERIMENTS AND RESULTS ON TARGET DATASET

First, we delete the feature columns with most missing data in the physical examination data set, leaving 32 features. Then, we use the average interpolation method to fill in the small part of the missing values. After data preprocessing, the data set is standardized, and then divided into training set and test set in a ratio of 8:2, 80% for training set and 20% for test set.

Due to the existence of redundant features, feature selection is required for the dataset. After we built the GA-RF algorithm model, we used the grid search method to determine the hyperparameters of the genetic algorithm, and sent the obtained feature subset to LightGBM for fitting and MSE evaluation, and used the better feature subset as the initial population to restart the selection of the genetic algorithm. After feature selection by the GAL algorithm, the attribute features of the target dataset were reduced from 32 to 18, which improved the efficiency of subsequent model training and prediction and obtained better prediction results.

For stacked models and deep neural networks, the training set sample size is small and there is an imbalance in the blood glucose range samples, so the training set needs to be oversampled and data enhanced. We first roughly divide the blood glucose distribution in the training set into three categories: hypoglycemia, normoglycemia, and hyperglycemia, and use KDE technology to estimate the probability distribution functions of hypoglycemia and hyperglycemia, then resample and combine with the original training set, and finally use GAN to perform data enhancement on the new training set to expand the training set. This combination of KDE and GAN can increase sample diversity while balancing data distribution, thereby reducing the risk of overfitting.

TABLE 3: Performance of different modules on the testing dataset.

| Model | MSE | MAE | R-square |
|---|---|---|---|
| LASSO | 1.740 | 0.889 | 0.135 |
| GA-LASSO | 1.694 | 0.850 | 0.158 |
| GKN-LASSO | 3.333 | 1.614 | -0.657 |
| CNN | 1.752 | 0.796 | 0.128 |
| GA-CNN | 1.669 | 0.794 | 0.171 |
| GKN-CNN | 1.805 | 0.916 | 0.102 |
| LSTM | 1.736 | 0.727 | 0.143 |
| GA-LSTM | 1.731 | 0.755 | 0.145 |
| GKN-LSTM | 1.654 | 0.750 | 0.179 |
| SVM | 1.687 | 0.717 | 0.161 |
| GA-SVM | 1.659 | **0.707** | 0.175 |
| GKN-SVM | 1.651 | 0.724 | 0.180 |
| RF | 1.623 | 0.786 | 0.193 |
| GA-RF | 1.598 | 0.785 | 0.206 |
| GKN-RF | 1.575 | 0.782 | 0.217 |
| LightGBM | 1.614 | 0.805 | 0.198 |
| GA-LightGBM | 1.579 | 0.799 | 0.215 |
| GKN-LightGBM | 1.549 | 0.825 | 0.230 |
| Stacking | 1.673 | 0.805 | 0.168 |
| GA-Stacking | 1.604 | 0.797 | 0.202 |
| **GKN-Stacking** | **1.529** | 0.798 | **0.242** |

For the stacked model, the number of layers and neurons of the basic model LSTM is determined by grid search to be two LSTM layers, with 36 neurons in the first layer and 18 neurons in the second layer. At the same time, the self-attention layer and residual link are introduced; the hyperparameters of RF are also determined by grid search. LightGBM has many hyperparameters, and Bayesian optimization is used to adjust the parameters.

Table 3 shows the performance comparison of each machine performance before and after the feature engineering method. Compared with basic models, the GA-RF feature selection algorithm can improve the performance of the model to a certain extent. Among them, CNN has grown the greatest. The MSE value was reduced from 1.752 to 1.669, and the improvement rate was 4.53%. Followed by the Stacking model, the MSE value was reduced from 1.673 to 1.604, and the improvement rate was 4.1%. Explain that the feature selection algorithm based on GA-RF can improve the predictive ability of the model.

After feature selection and data enhancement technology, except for lasso and CNN, the performance of the remaining models has been greatly improved compared with basic models. LSTM's MSE increased from 1.736 to 1.654, and R square increased from 0.143 to 0.179; SVM's MSE decreased from 1.687 to 1.651, R-Square to 0.217; LightGBM's MSE decreased from 1.614 to 1.549, R square rose from 0.198 to 0.230 ; Stack MSE decreased from 1.673 to 1.529, and R-Square increased from 0.168 to 0.242. It shows that in most



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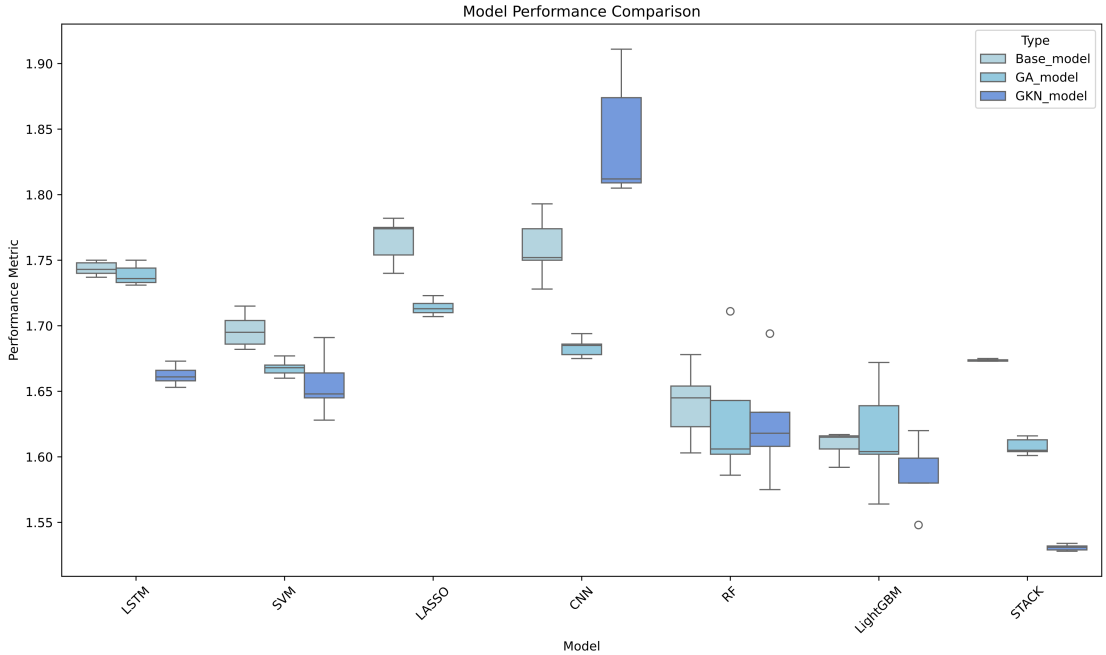



FIGURE 8: MSE box plots for different forecasting methods.

cases, using KDE-GAN's data enhancement technology will bring performance improvement, which can better cope with the imbalance of the sample class, improve the effect of data enhancement, thereby reducing the risk of excessive fitting and improving the risk Model generalization ability.

Figure 8 shows the K-FOLD cross-validation results of each experiment. As can be seen from the box plot, the stacking model (1.529) performs best in terms of MSE. In R-Square, the Ganges cumulative model (0.242) also performs best and has high stability. GKN-LASSO performs worst. Therefore, the Ganges cumulative model performs best on most evaluation indicators. The smaller the entity of the line graph, the higher the stability of the model, and the box entity of Stack is the smallest in the three cases, indicating that the stacking model can improve the stability of the model. In addition, in most cases, the combination of genetic algorithm and GAN processing method (GKN) improves performance. This shows that this method has potential when processing a given data set, but it may lead to performance degradation in some cases. Therefore, detailed evaluation and selection are required in practical applications.

In terms of selecting sub-learners, LSTM performs well in capturing complex feature dependencies and patterns, and is suitable for data with potentially complex relationships. The significant improvement of GKN-LSTM shows that LSTM has great potential when combined with other optimization techniques. SVM performs well when processing high-dimensional data and fewer samples, and is suitable for data sets with complex features and high dimensions. Its MAE value performs best among all machine learning algorithms. RF improves the stability and robustness of the model by integrating multiple decision trees. The significant improvement of GA-RF and GKN-RF shows that RF can effectively use other optimization techniques in integrated learning to further improve performance. LightGBM has the best performance, and the performance is significantly improved after integration, which can improve the overall prediction performance and generalization ability of the model. Therefore, this paper makes full use of the respective advantages of LSTM, SVM, RF and LightGBM. This diversity and complementarity are the main reasons for choosing them as sub-learners for ensemble learning.

In summary, the stacking model proposed in this article has strong prediction performance after feature engineering. The MAE is 1.529, MAE is 0.798, and R square is 0.242. Except for MAE, other indicators are better than other traditional machine learning. The algorithm has increased by more than 1 %, indicating that the stacking model proposed in this article is more suitable for blood glucose forecasting.

### C. EXPERIMENTS AND RESULTS ON PIMA INDIANS DIABETES AND FUSED DATASET

There is often a low correlation and a highly nonlinear relationship between routine blood test data and blood glucose levels. Capturing the relationship between each feature and blood glucose label becomes a complex challenge. The R-squared performance of each model is low, making it difficult to fully demonstrate the predictive ability of the model.

For the classification problem, in order to further verify the generalization ability and practical application effect of



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TABLE 4: Performance of different modules on the PIMA Indian dataset.

| Ref | Model | Accuracy |
|---|---|---|
| [13] | LTC | 85% |
| [38] | LSTM | 87.26% |
| [39] | LGBM | 92.5% |
| [16] | CNN-BiLSTM | 98% |
| [40] | DNN | 86.26% |
| [41] | Tri-Ensemble Model | 97.49% |
| **Proposed** | **GKN-Stack** | **98.96%** |

stacking proposed in this paper on other data sets, we selected the classic PIMA Indian diabetes data set and its fusion data set as the target data set. We first preprocessed the data, including encoding, discretization, normalization, and data set partitioning. After data standardization, feature selection, data enhancement, etc., the GKN-Stack model designed in this paper was used for training classification. Table 4 lists the accuracy of the models that currently perform well on the PIMA Indian data set. It can be seen that the accuracy of the model proposed in this paper reaches 98.96%, which is better than the existing models.

Table 5 shows the performance of machine learning algorithms such as logistic regression, SVM, KNN, random forest and GKN-Stack on the fusion data set. In terms of accuracy, the GKN-Stacking model proposed in this paper has improved compared with traditional machine learning algorithms, and its evaluation indicators are better than other algorithms. Acc, Pre, Rec, and FM reached 99.09%, 99.62%, 97.76%, and 98.68% respectively, and RMSE and MAE also reached excellent performance of 0.09 and 0.01 respectively, indicating that the performance of the GKN-stack model in diabetes prediction is much better than that of traditional machine learning algorithms.

In summary, compared with traditional machine learning algorithms such as KNN, SVM, LR, and RF, the proposed GKN-Stack achieved better accuracy, and other indicators were also better than other algorithms. It is a good diabetes risk prediction method and is suitable for early diabetes risk prediction in patients.

## V. DISCUSSION AND CONCLUSION

### A. DISCUSSION

Although the model proposed in this paper performs better in regression and classification problems than traditional non-computer learning algorithms, there are certain limitations in predicting blood glucose using routine blood test data because there is high complexity and low correlation between routine blood test data and blood glucose levels. The feature correlation heat map shown in Figure 3 also shows that the correlation between routine blood test data and blood glucose levels is very low, which becomes a major challenge in predicting blood glucose levels. We believe that the specific factors that cause this are as follows:

- The correlation between conventional blood detection parameters and blood glucose levels is generally low. This inner difficulty is that blood glucose is affected by various factors, including diet, physical activity, pressure, and drugs. These factors cannot be directly captured by conventional blood testing, making it a noise and affecting the real value of the physical examination data and the real value of blood glucose Between correlation. This makes the MSE value of traditional machine learning algorithms on the forecast indicators of the medical examination dataset.
- In addition, conventional blood testing is usually performed at a certain time, which may not reflect the short-term fluctuations of blood glucose levels. This time limit reduces the accuracy of prediction based on these data.
- The reaction to various factors affecting blood glucose is largely different. For example, insulin capabilities of hereditary, lifestyle, lifestyle, complications, and secretion capabilities will seriously affect blood glucose levels, which is a huge challenge for the prediction of creating a model suitable for everyone.

In order to cope with these limitations, future research should pay attention to combining conventional blood detection data with other related data sources, such as continuous blood glucose monitoring, diet records and physical activity tracking. The combination of many data flow can improve the accuracy of prediction and provide a more comprehensive view of individual blood glucose regulation.

### B. CONCLUSION

Good blood glucose management is very important for preventing the occurrence of diabetes and reducing complications. This paper proposes a GKN-Stack model for physiological data to predict blood glucose and diabetes classification. In view of the characteristics of high-dimensional data set samples, a GA feature selection algorithm based on random forest under the LightGBM evaluation strategy is proposed to ensure the optimality of the attribute set. At the same time, the smooth diversity GAN of KDE is used to effectively alleviate the problems of unbalanced sample intervals and small sample size of the data set. Experiments show that the method combining KDE and GAN can achieve the best effect, significantly improving the performance indicators of the model, and the blood glucose prediction return of the stack model is better than other machine learning algorithms in terms of MSE and R evaluation indicators, reaching 1.529 and 0.242 respectively. Finally, the proposed stacking model also performs well on the classic PIMA Indian diabetes classification dataset. Indicators such as ACC and PER have reached 97.0%, and the performance is far better than other traditional machine learning algorithms. Although there is a significant challenge to predict blood glucose from conventional blood testing data, it is still a research field with potential clinical interests. Acknowledge these limitations and seek solutions,



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TABLE 5: Performance of different modules on the fused dataset.

| Classifier model | Acc (%) | Pre (%) | Rec (%) | FM (%) | RMSE | MAE |
|---|---|---|---|---|---|---|
| Naive Bayes | 74.01 | 64.00 | 55.00 | 59.00 | 0.26 | 0.51 |
| Logistic Regression | 76.89 | 73.23 | 49.73 | 59.23 | 0.48 | 0.23 |
| SVM | 76.71 | 71.01 | 52.40 | 60.31 | 0.48 | 0.23 |
| Decision tree | 84.30 | 77.00 | 78.00 | 77.00 | 0.16 | 0.40 |
| K-nearest neighbors | 85.56 | 76.88 | 81.81 | 79.27 | 0.38 | 0.14 |
| Random forest | 97.83 | 98.34 | 95.18 | 96.73 | 0.15 | 0.02 |
| **GKN-Stack** | **99.09** | **99.62** | **97.76** | **98.68** | **0.09** | **0.01** |

such as data integration and personalized methods, can pave the way for more reliable and practical blood glucose management applications.

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